Decomposing Exchange Rate Responses to News: Evidence from South Africa

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1 Introduction

Coleman and Karagedikli (2012)

2 Literature review

3 Data and methodology

3.1 Data

This study draws on four data sources: SA monetary policy surprises from Pirozhkova et al. (2024), South African economic data releases from **add**, ZARONIA overnight index swaps (OIS) from the South African Reserve Bank (SARB), and US monetary policy surprises from Jarociński (2024). The sample runs from April 2008 to November 2024, with the end date determined by the availability of ZARONIA OIS data. All SARB announcement days within this period are included, with limited exceptions where observations were dropped due to missing values or release dates that did not coincide with announcement days.

The SA monetary policy surprises are summarised in Figure A1. A vast literature exists on the high frequency identification of exogenous monetary policy shocks (see Kuttner, 2001; Cook and Hahn, 1989; Gurkaynak et al., 2005; Swanson, 2021; Miranda-Agrippino and Ricco, 2021). Pirozhkova et al. (2024) follow in this tradition isolating four surprises from high frequency South African financial market data. These are the traditional Target, Forward Guidance (FG) surprises, and the less traditional Central Bank Information (CBI), and Country Risk (CR) surprises. The target surprise account for the immediate response by investors to a policy announcement, whilst forward guidance incorporates responses to the communicated longer term path of policy. The CBI can be viewed as a subset of the FG, which specifically focuses on the economy (Jarociński, 2024). The CR surprise is a novel addition by Pirozhkova et al. (2024), which details the role of the central bank in responding to external shocks in a small open economy, such as South Africa. For our purposes, these surprises provide the best available estimates of monetary policy surprises in South Africa for analysing the exchange

rate and interest rate response to monetary policy surprises. As a supplement our analysis of the response to South African news, by using data on economic release - cpi, gdp, and unemployment. The surprise to these news is defined as the difference between the actual release and the forecasts from prominent analysts. To understand the equivalent response to US surprises, we made use of the Jarociński (2024) surprises (see Figure A2). These are defined exactly as the Pirozhkova et al. (2024), with the exclusion of the CR surprise, and the inclusion of the Large-scale Asset Purchases (LSAP) surprise. Lastly, the SARB recently undertook a reform of its benchmark rate the JIBAR. The result of that process is the ZARONIA OIS data, which is used to measure overnight interest rates in South Africa (see Figure A3). The 2, 5, and 10 year tenors are of interest to us. These give a detailed picture of the term structure of interest rates and their response to monetary policy shocks. The key variables for the analysis are shown in Figure A4

3.2 Methodology

4 Results

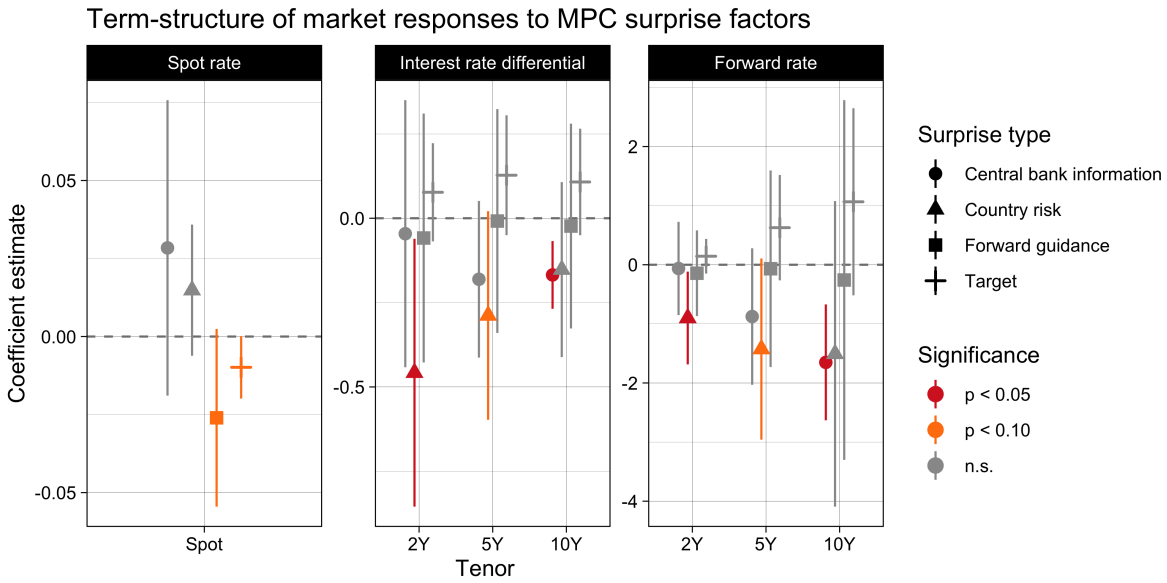


Figure 1: Response to Monetary Policy Surprises

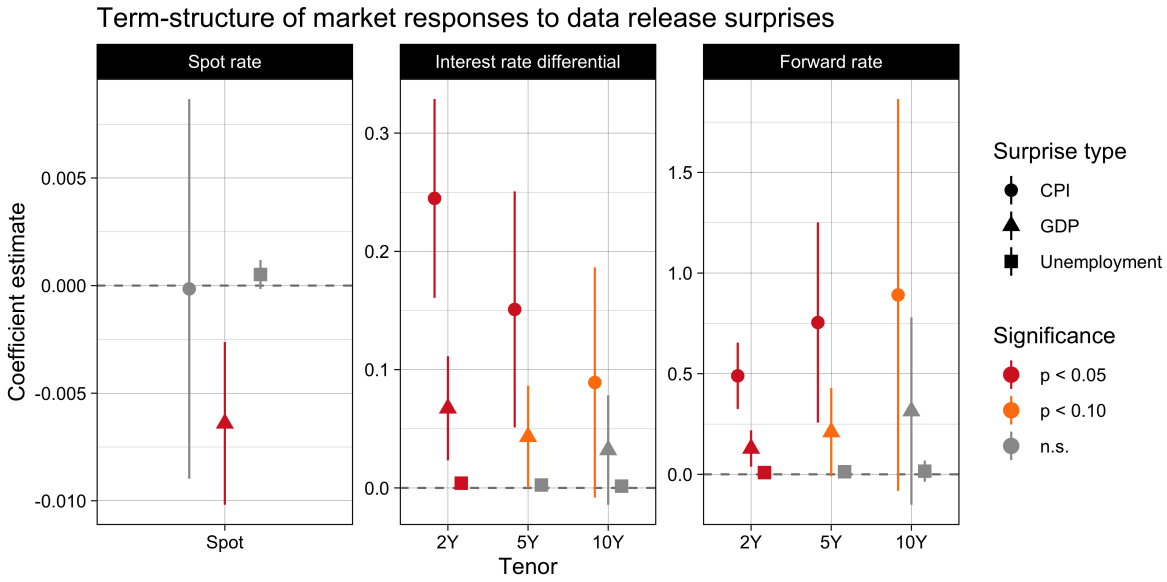


Figure 2: Data release response

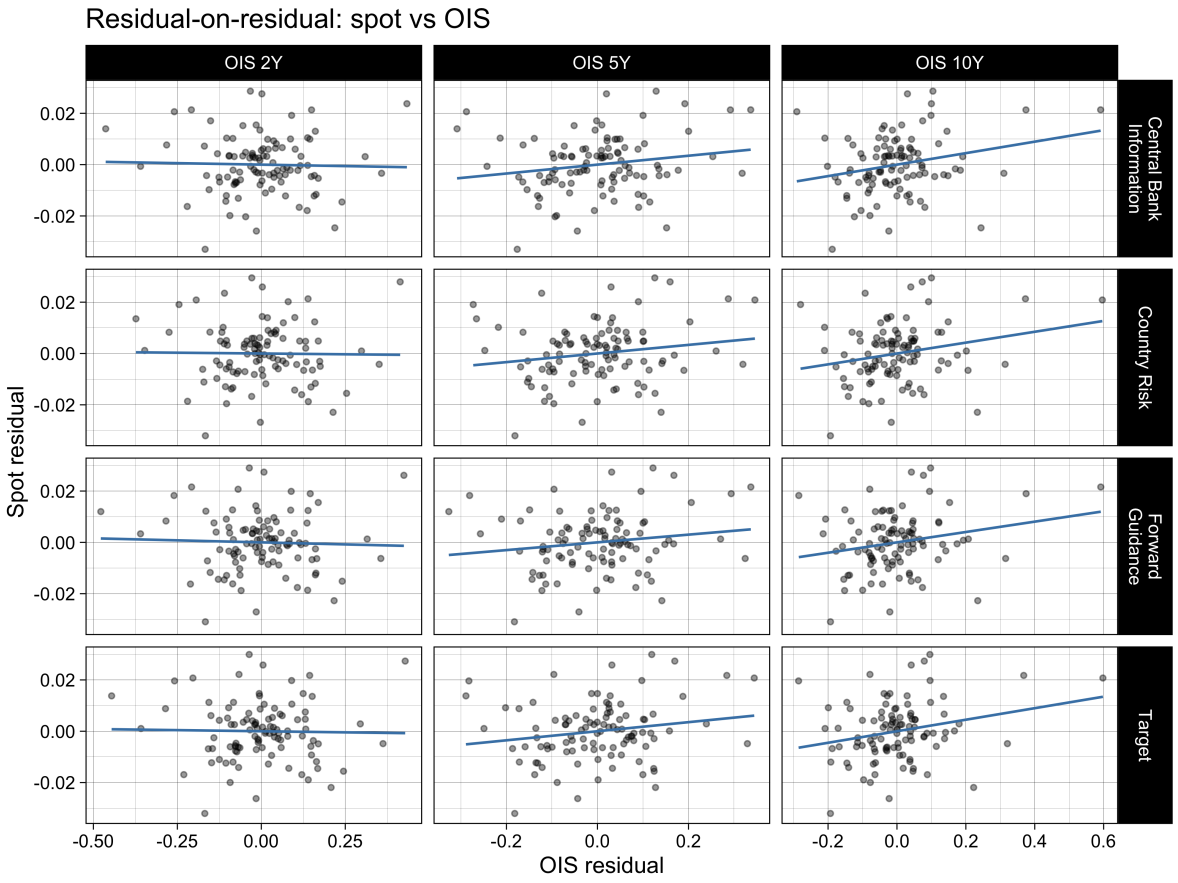


Figure 3: Spot residuals versus interest rate residual

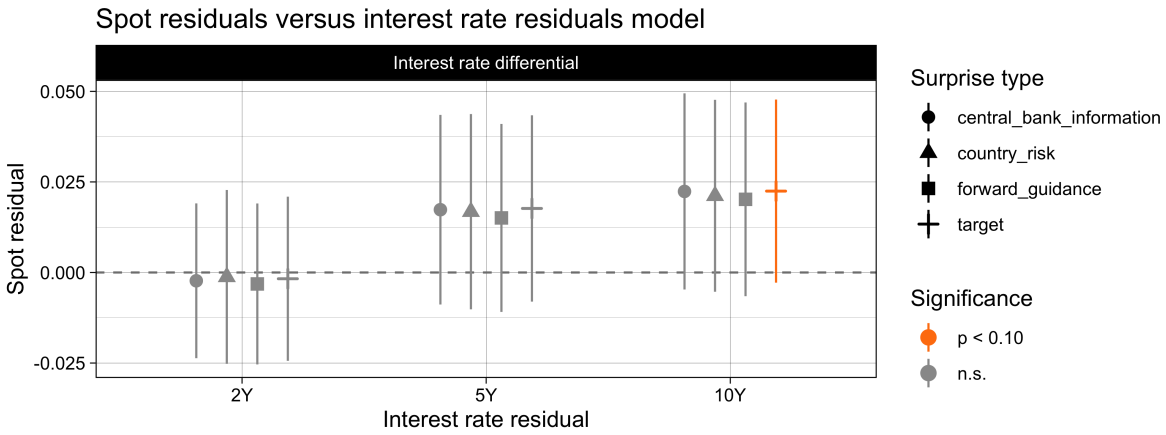


Figure 4: Spot residuals versus interest rate residuals model

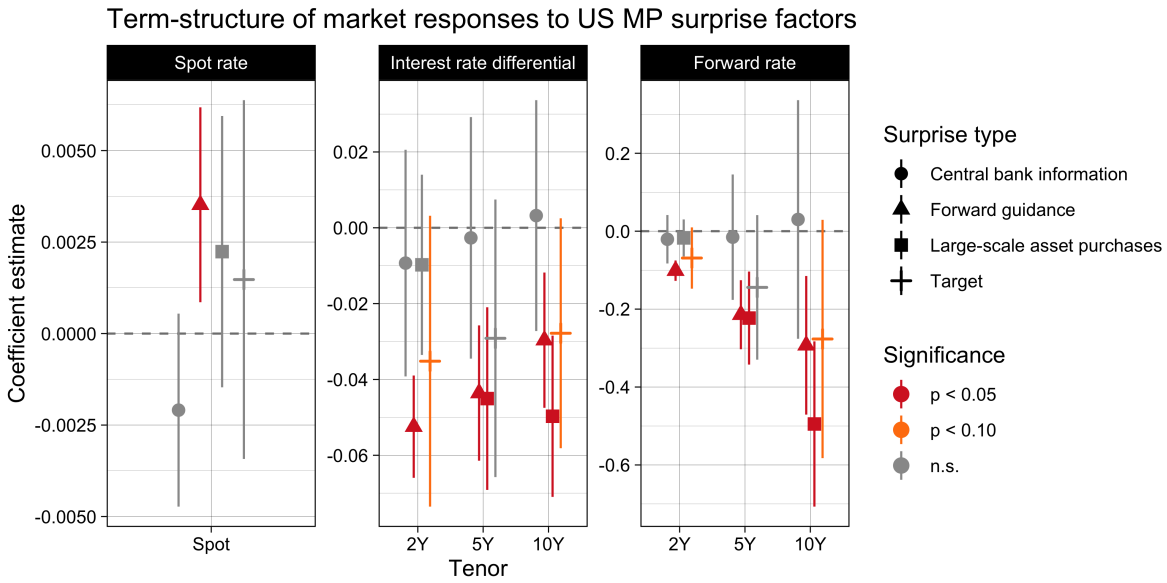


Figure 5: US MPS response

5 Conclusion

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A Appendix

A.1 Data

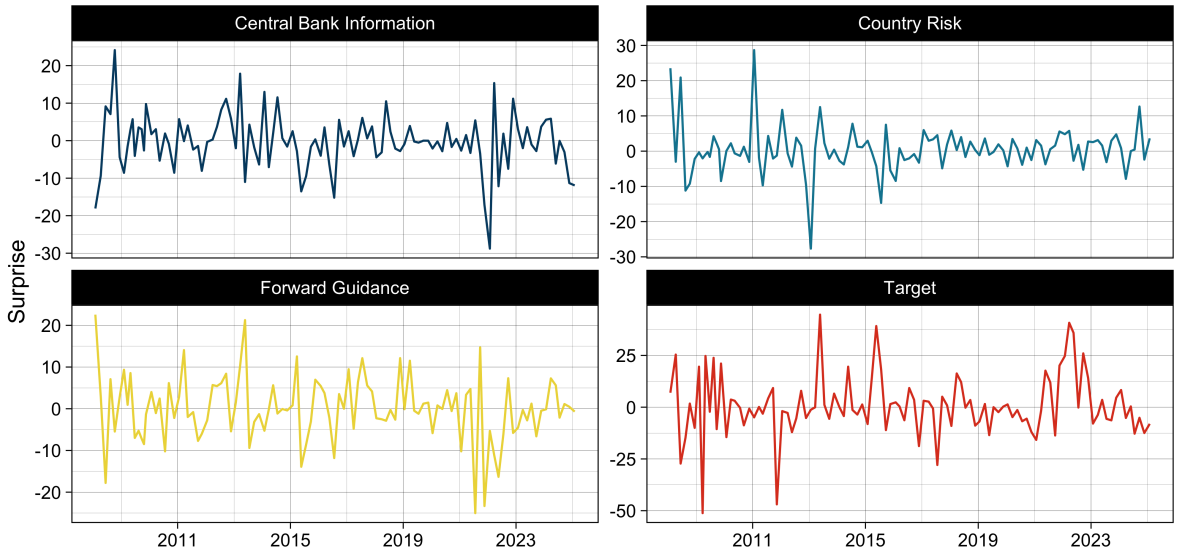


Figure A1: SA monetary policy surprises

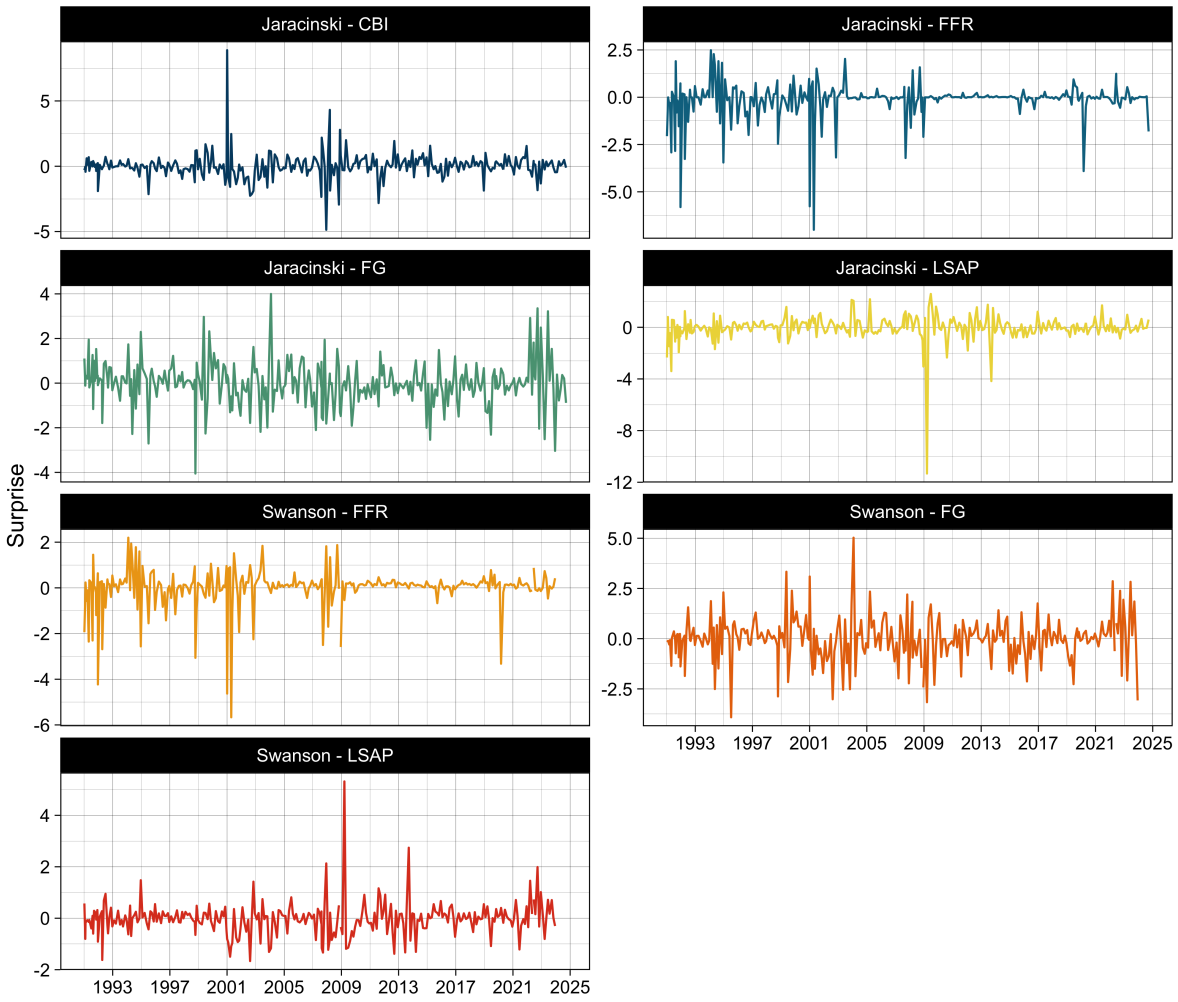


Figure A2: US monetary policy surprises

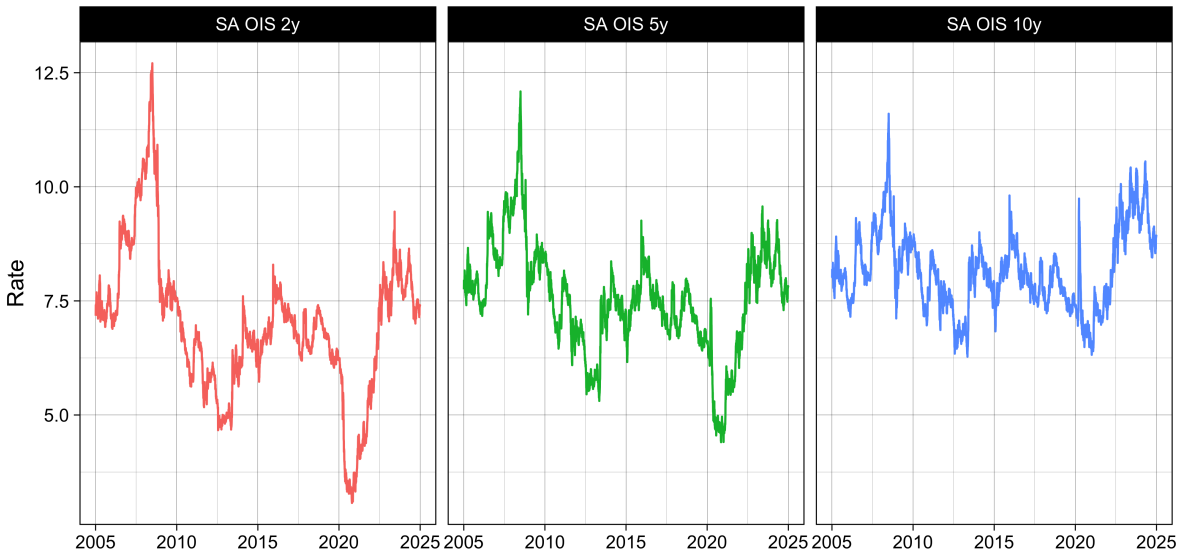


Figure A3: SA ZARONIA Overnight Index Rates

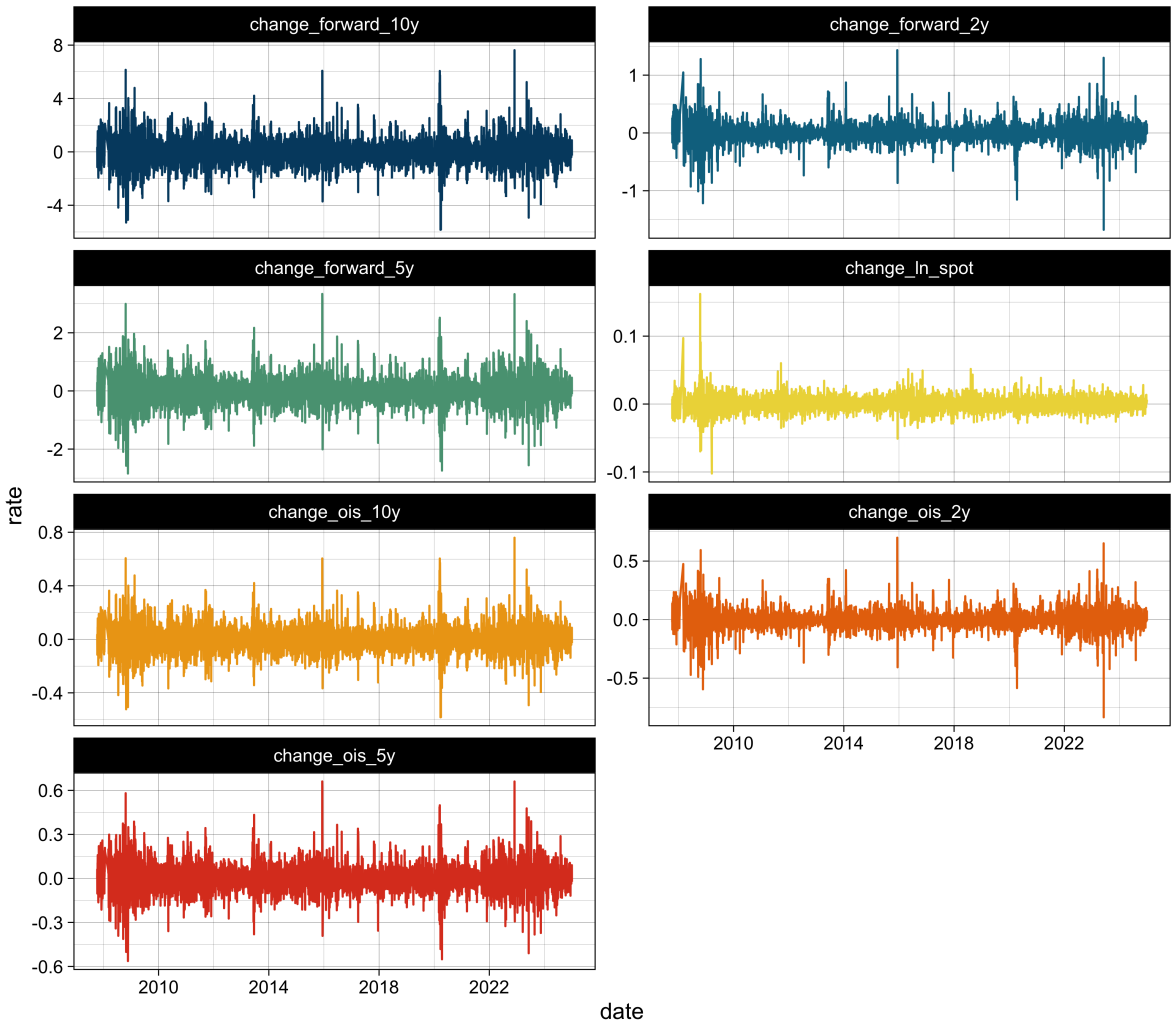


Figure A4: Key variables

Table A1: Descriptive statistics

series	Mean	Median	SD	Min	Max	Observations
change_forward_10y	0.0029	-0.0155	1.0217	-5.8376	7.6202	4,137
change_forward_2y	-0.0007	-0.0059	0.1757	-1.6754	1.4316	4,137
change_forward_5y	-0.0005	-0.0098	0.4990	-2.8409	3.3358	4,137
change_ln_spot	0.0002	0.0000	0.0114	-0.1023	0.1621	4,137
change_ois_10y	0.0003	-0.0017	0.1018	-0.5827	0.7593	4,137
change_ois_2y	-0.0005	-0.0028	0.0862	-0.8343	0.6997	4,137
change_ois_5y	-0.0002	-0.0021	0.0990	-0.5631	0.6607	4,137

A.2 Detailed results

Table A2: SA monetary policy surprises models

surprise type	dependant variables	estimate	std.error	statistic	p.value	r.squared	nobs
target models							
	change_ln_spot	-0.010	0.005	-1.931	0.056	0.017	101
	change_ois_2y	0.077	0.074	1.037	0.302	0.007	101
	change_ois_5y	0.127	0.091	1.407	0.163	0.027	101
	change_ois_10y	0.108	0.081	1.335	0.185	0.018	101
	change_forward_2y	0.144	0.150	0.963	0.338	0.007	101
	change_forward_5y	0.628	0.455	1.379	0.171	0.026	101
	change_forward_10y	1.065	0.807	1.320	0.190	0.017	101
forward guidance models							
	change_ln_spot	-0.026	0.015	-1.793	0.076	0.030	101
	change_ois_2y	-0.059	0.188	-0.311	0.756	0.001	101
	change_ois_5y	-0.008	0.169	-0.049	0.961	0.000	101
	change_ois_10y	-0.023	0.155	-0.150	0.881	0.000	101
	change_forward_2y	-0.143	0.370	-0.387	0.699	0.002	101
	change_forward_5y	-0.068	0.848	-0.080	0.937	0.000	101
	change_forward_10y	-0.258	1.552	-0.166	0.868	0.000	101
central bank information models							
	change_ln_spot	0.028	0.024	1.176	0.242	0.033	101
	change_ois_2y	-0.046	0.202	-0.227	0.821	0.001	101
	change_ois_5y	-0.181	0.118	-1.527	0.130	0.012	101
	change_ois_10y	-0.168	0.051	-3.281	0.001	0.010	101
	change_forward_2y	-0.063	0.403	-0.157	0.875	0.000	101
	change_forward_5y	-0.876	0.589	-1.487	0.140	0.012	101
	change_forward_10y	-1.652	0.500	-3.303	0.001	0.010	101
country risk models							
	change_ln_spot	0.015	0.011	1.386	0.169	0.007	101
	change_ois_2y	-0.458	0.202	-2.261	0.026	0.047	101
	change_ois_5y	-0.288	0.158	-1.828	0.071	0.024	101
	change_ois_10y	-0.152	0.132	-1.151	0.253	0.006	101
	change_forward_2y	-0.901	0.400	-2.250	0.027	0.045	101
	change_forward_5y	-1.426	0.782	-1.824	0.071	0.023	101
	change_forward_10y	-1.507	1.319	-1.143	0.256	0.006	101

Table A3: SA residuals models

surprise type	predictor_model	term	estimate	std.error	statistic	p.value	r.squared	nobs
central bank information models								
	change_ois_2y	resid_predictor	-0.002	0.011	-0.210	0.834	0.001	101
	change_ois_5y	resid_predictor	0.017	0.013	1.298	0.197	0.033	101
	change_ois_10y	resid_predictor	0.022	0.014	1.620	0.108	0.058	101
country risk models								
	change_ois_2y	resid_predictor	-0.001	0.012	-0.100	0.921	0.000	101
	change_ois_5y	resid_predictor	0.017	0.014	1.221	0.225	0.029	101
	change_ois_10y	resid_predictor	0.021	0.014	1.567	0.120	0.050	101
forward guidance models								
	change_ois_2y	resid_predictor	-0.003	0.011	-0.279	0.781	0.001	101
	change_ois_5y	resid_predictor	0.015	0.013	1.138	0.258	0.025	101
	change_ois_10y	resid_predictor	0.020	0.014	1.480	0.142	0.047	101
target models								
	change_ois_2y	resid_predictor	-0.002	0.012	-0.150	0.881	0.000	101
	change_ois_5y	resid_predictor	0.018	0.013	1.347	0.181	0.033	101
	change_ois_10y	resid_predictor	0.022	0.013	1.742	0.085	0.057	101

Table A4: SA data release surpises models

surprise type	dependant variables	estimate	std.error	statistic	p.value	r.squared	nobs
cpi models							
	change_ln_spot	-0.000	0.001	-0.154	0.878	0.000	202
	change_ln_spot	-0.000	0.005	-0.034	0.973	0.000	202
	change_ois_2y	-0.003	0.005	-0.619	0.537	0.194	202
	change_ois_2y	0.245	0.043	5.708	0.000	0.194	202
	change_ois_5y	-0.012	0.007	-1.789	0.075	0.058	202
	change_ois_5y	0.151	0.051	2.964	0.003	0.058	202
	change_ois_10y	-0.014	0.007	-1.881	0.061	0.019	202
	change_ois_10y	0.089	0.050	1.795	0.074	0.019	202
	change_forward_2y	-0.006	0.010	-0.621	0.536	0.190	202
	change_forward_2y	0.489	0.084	5.810	0.000	0.190	202
	change_forward_5y	-0.062	0.035	-1.789	0.075	0.058	202
	change_forward_5y	0.754	0.254	2.971	0.003	0.058	202
	change_forward_10y	-0.140	0.075	-1.880	0.062	0.019	202
	change_forward_10y	0.891	0.497	1.794	0.074	0.019	202
gdp models							
	change_ln_spot	-0.001	0.002	-0.410	0.683	0.086	68
	change_ln_spot	-0.006	0.002	-3.320	0.001	0.086	68
	change_ois_2y	0.003	0.012	0.206	0.837	0.113	68
	change_ois_2y	0.067	0.022	2.998	0.004	0.113	68
	change_ois_5y	0.017	0.013	1.312	0.194	0.079	68
	change_ois_5y	0.043	0.022	1.958	0.054	0.079	68
	change_ois_10y	0.026	0.012	2.082	0.041	0.038	68
	change_ois_10y	0.032	0.024	1.357	0.179	0.038	68
	change_forward_2y	0.004	0.026	0.165	0.869	0.100	68
	change_forward_2y	0.128	0.046	2.784	0.007	0.100	68
	change_forward_5y	0.083	0.065	1.278	0.206	0.073	68
	change_forward_5y	0.210	0.112	1.878	0.065	0.073	68
	change_forward_10y	0.255	0.124	2.059	0.043	0.036	68
	change_forward_10y	0.314	0.237	1.323	0.190	0.036	68
south africa unemployment models							
	change_ln_spot	-0.001	0.002	-0.570	0.571	0.008	49
	change_ln_spot	0.001	0.000	1.495	0.142	0.008	49
	change_ois_2y	0.009	0.008	1.124	0.267	0.019	49
	change_ois_2y	0.004	0.002	2.268	0.028	0.019	49
	change_ois_5y	0.012	0.010	1.225	0.227	0.004	49
	change_ois_5y	0.002	0.002	1.144	0.258	0.004	49
	change_ois_10y	0.008	0.011	0.796	0.430	0.001	49
	change_ois_10y	0.002	0.003	0.567	0.574	0.001	49
	change_forward_2y	0.018	0.017	1.029	0.309	0.020	49
	change_forward_2y	0.009	0.004	2.357	0.023	0.020	49
	change_forward_5y	0.060	0.050	1.190	0.240	0.004	49
	change_forward_5y	0.013	0.011	1.167	0.249	0.004	49
	change_forward_10y	0.083	0.106	0.782	0.438	0.001	49
	change_forward_10y	0.016	0.027	0.581	0.564	0.001	49

Table A5: US monetary policy surprises models

surprise type	dependant variables	estimate	std.error	statistic	p.value	r.squared	nobs
target models							
	change_ln_spot	0.001	0.003	0.590	0.556	0.003	123
	change_ois_2y	-0.035	0.020	-1.798	0.075	0.028	123
	change_ois_5y	-0.029	0.019	-1.561	0.121	0.020	123
	change_ois_10y	-0.028	0.015	-1.798	0.075	0.020	123
	change_forward_2y	-0.069	0.040	-1.717	0.088	0.027	123
	change_forward_5y	-0.144	0.095	-1.524	0.130	0.019	123
	change_forward_10y	-0.277	0.156	-1.773	0.079	0.020	123
forward guidance models							
	change_ln_spot	0.004	0.001	2.591	0.011	0.067	123
	change_ois_2y	-0.052	0.007	-7.620	0.000	0.252	123
	change_ois_5y	-0.044	0.009	-4.787	0.000	0.176	123
	change_ois_10y	-0.030	0.009	-3.257	0.001	0.091	123
	change_forward_2y	-0.101	0.013	-7.617	0.000	0.233	123
	change_forward_5y	-0.214	0.045	-4.744	0.000	0.171	123
	change_forward_10y	-0.293	0.091	-3.227	0.002	0.089	123
central bank information models							
	change_ln_spot	-0.002	0.001	-1.555	0.123	0.020	123
	change_ois_2y	-0.009	0.015	-0.611	0.543	0.007	123
	change_ois_5y	-0.003	0.016	-0.164	0.870	0.001	123
	change_ois_10y	0.003	0.016	0.207	0.836	0.001	123
	change_forward_2y	-0.021	0.032	-0.653	0.515	0.008	123
	change_forward_5y	-0.015	0.082	-0.187	0.852	0.001	123
	change_forward_10y	0.030	0.156	0.192	0.848	0.001	123
lsap models							
	change_ln_spot	0.002	0.002	1.184	0.239	0.014	123
	change_ois_2y	-0.010	0.012	-0.806	0.422	0.005	123
	change_ois_5y	-0.045	0.012	-3.666	0.000	0.097	123
	change_ois_10y	-0.050	0.011	-4.581	0.000	0.133	123
	change_forward_2y	-0.017	0.024	-0.715	0.476	0.003	123
	change_forward_5y	-0.223	0.061	-3.661	0.000	0.095	123
	change_forward_10y	-0.495	0.108	-4.574	0.000	0.131	123